Microsoft Elevate

CAPSTONE PROJECT

Healthcare Infrastructure Analytics of AIIMS Hospitals in India Using Power BI

PRESENTED BY

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OUTLINE:

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PROBLEM STATEMENT:

- Healthcare infrastructure plays a crucial role in delivering quality medical services. Government hospitals such as AIIMS must effectively manage and monitor critical resources such as ICU beds, oxygen beds, and operational hospital beds. However, analyzing raw infrastructure data across multiple hospitals can be difficult and time-consuming. Without proper visualization, it becomes challenging for administrators to identify hospital capacity, compare infrastructure levels, and make data-driven decisions.

Therefore, there is a need for an analytical dashboard that can visually present healthcare infrastructure data for better understanding and decision-making.

PROPOSED SOLUTION:

- The proposed solution is to develop a **Power BI dashboard** that analyzes healthcare infrastructure data of AIIMS hospitals across India.
- The dashboard provides:
- Summary metrics of ICU beds, oxygen beds, and operational beds
- Hospital-wise infrastructure comparison
- Distribution of critical care and oxygen bed capacity
- Interactive filtering for individual hospitals
- Visual insights to support healthcare infrastructure planning
- This dashboard helps stakeholders easily analyze and interpret hospital infrastructure data.

SYSTEM APPROACH:

- The system was developed using **Microsoft Power BI**, a powerful data visualization and business intelligence tool.

Tools and Technologies Used

- **Microsoft Power BI Desktop**
- **Power Query** for data transformation and cleaning
- **DAX (Data Analysis Expressions)** for creating calculated measures
- **Data Visualization techniques** such as bar charts, donut charts, and KPI cards

System Requirements

- Power BI Desktop
- Operationalization of dedicated hospital beds, and beds with ventilators dataset
- Windows operating system

DASHBOARD DEVELOPMENT PROCESS:

The following steps were followed to develop the dashboard:

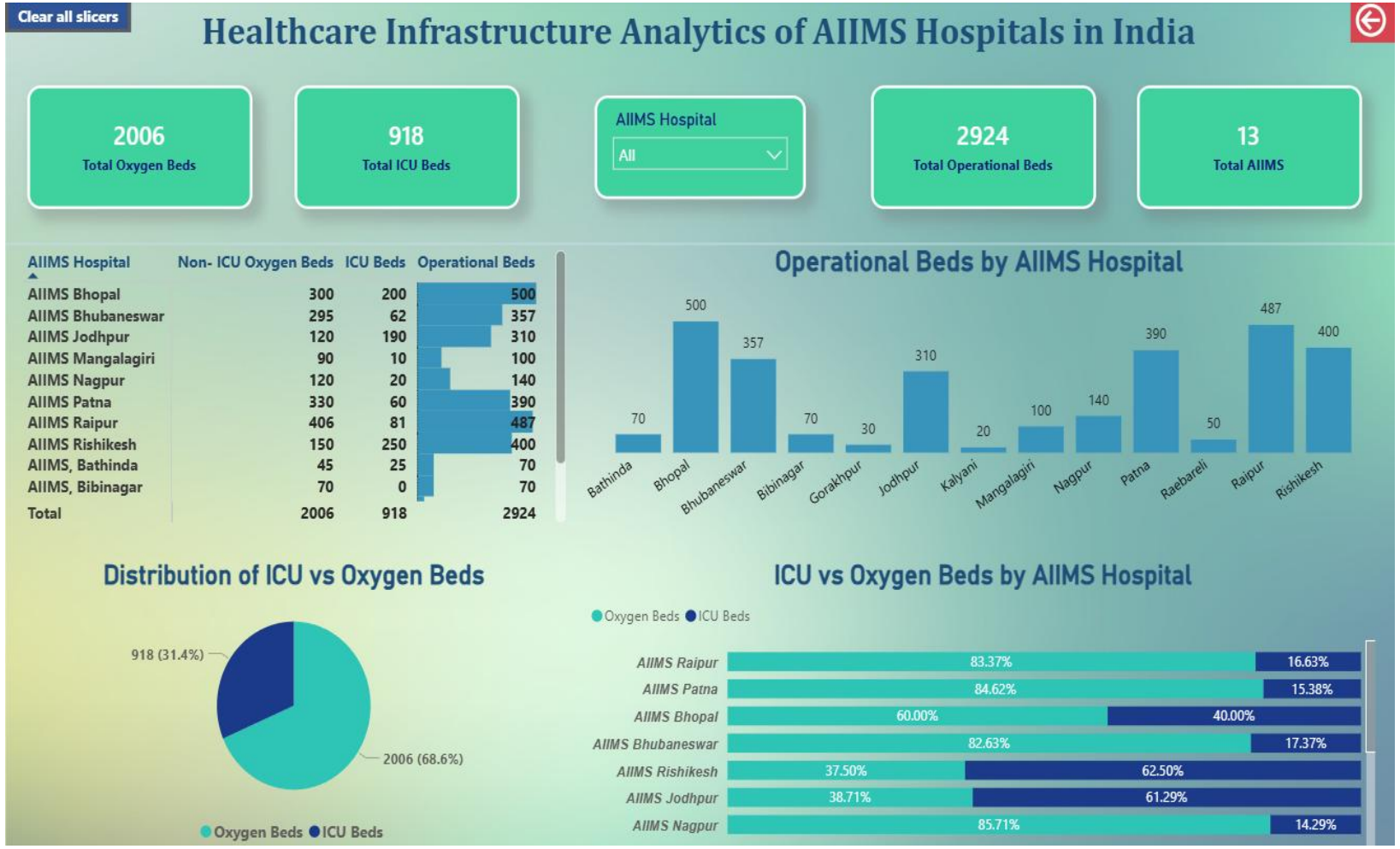
1. Import the Operationalization of dedicated hospital beds, and beds with ventilators into Power BI.
2. Perform data cleaning and transformation using **Power Query**.
3. Create calculated measures using **DAX functions** to compute totals and key metrics.
4. Design interactive visualizations including:
    1. KPI cards
    2. Bar charts
    3. Donut charts
    4. Data tables
5. Add slicers to enable interactive filtering by hospital.
6. Arrange the visuals to create an interactive analytical dashboard.

RESULT:

The developed dashboard successfully visualizes the healthcare infrastructure data of AIIMS hospitals in India.

The dashboard provides:

- Total oxygen beds: **2006**
- Total ICU beds: **918**
- Total operational beds: **2924**
- Total AIIMS hospitals analyzed: **13**
- The dashboard enables users to compare hospital capacities, analyze bed distribution, and identify hospitals with higher healthcare infrastructure availability.



AIIMS Hospital	Non- ICU Oxygen Beds	ICU Beds	Operational Beds
AIIMS Bhopal	300	200	500
AIIMS Bhubaneswar	295	62	357
AIIMS Jodhpur	120	190	310
AIIMS Mangalagiri	90	10	100
AIIMS Nagpur	120	20	140
AIIMS Patna	330	60	390
AIIMS Raipur	406	81	487
AIIMS Rishikesh	150	250	400
AIIMS, Bathinda	45	25	70
AIIMS, Bibinagar	70	0	70
Total	2006	918	2924

CONCLUSION:

- The Power BI dashboard provides a clear and interactive visualization of healthcare infrastructure data across AIIMS hospitals. By transforming raw data into meaningful insights, the dashboard helps stakeholders understand hospital capacity, resource distribution, and infrastructure availability.
- This analytical approach improves data interpretation and supports better planning and management of healthcare infrastructure.

FUTURE SCOPE:

The project can be further enhanced by:

- Integrating **real-time hospital infrastructure data**
- Expanding the dataset to include **more hospitals and states**
- Adding **time-series analysis** to track infrastructure growth
- Incorporating **patient admission and bed occupancy data**
- Developing predictive models to forecast healthcare demand

REFERENCES:

- Microsoft Power BI Documentation
- The National Data and Analytics Platform (NDAP) - https://ndap.niti.gov.in/
- DATASET = Operationalization of dedicated hospital beds, and beds with ventilators - https://ndap.niti.gov.in/dataset/1358
- AICTE Internship Program Resources
- Power BI Report Link (Github) - https://github.com/alwinsudheesh/AIIMS-Healthcare-Infrastructure-PowerBI-Dashboard/blob/main/AIIMS_Healthcare_Infrastructure_Dashboard.pbix

Thank You